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Shipley

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(54) **FLOOD PANEL BARRIER SYSTEM FOR A THROUGHWAY OF A STRUCTURE**

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E06B 9/00 (2006.01)

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CPC ***E06B 9/00*** (2013.01); *E06B 2009/007* (2013.01)

(58) **Field of Classification Search**
CPC ... E04H 9/145; E02B 7/54; E02B 7/38; E02B 7/20; E02B 7/02; E02B 7/00; E06B 9/00; E06B 2009/015
See application file for complete search history.

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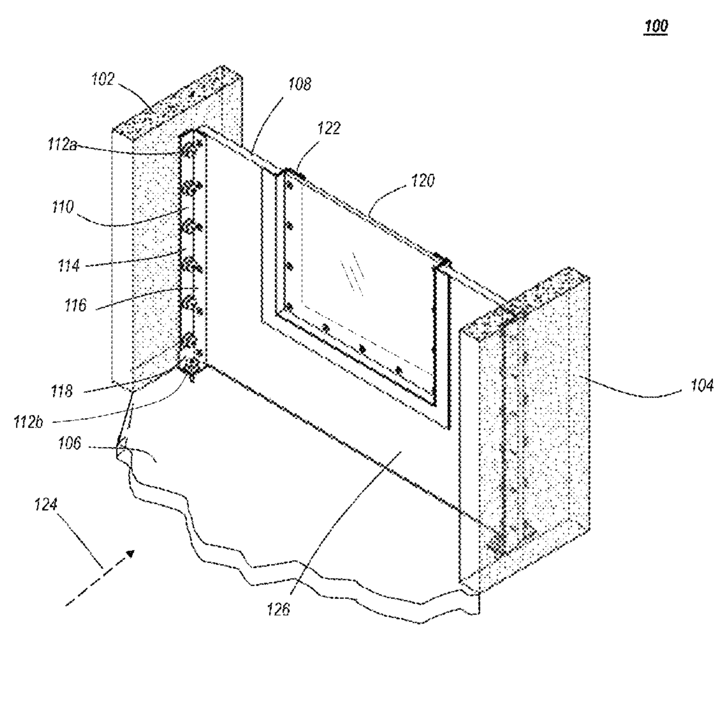
(57) **ABSTRACT**

A flood panel barrier system is configured to block flood water from intruding into a structure at a throughway that leads into the structure. The flood panel barrier system takes advantage of the walls of the structure being water impermeable, such that water cannot pass into the structure through the walls. A flood panel extends across the through way and is fastened to the opposing walls. Vertical angled brackets are used to fasten the flood panel to the walls, including gaskets between the bracket and the walls and flood panel. There is also a horizontal angled bracket across the bottom of the flood panel to fasten the flood panel to the ground surface, with a gasket between the ground surface and the horizontal angled bracket and the bottom of the flood panel. The flood panel has a removable portion that allows people to move through the flood panel when flood panel barrier system is installed, and before the are flood waters.

15 Claims, 9 Drawing Sheets



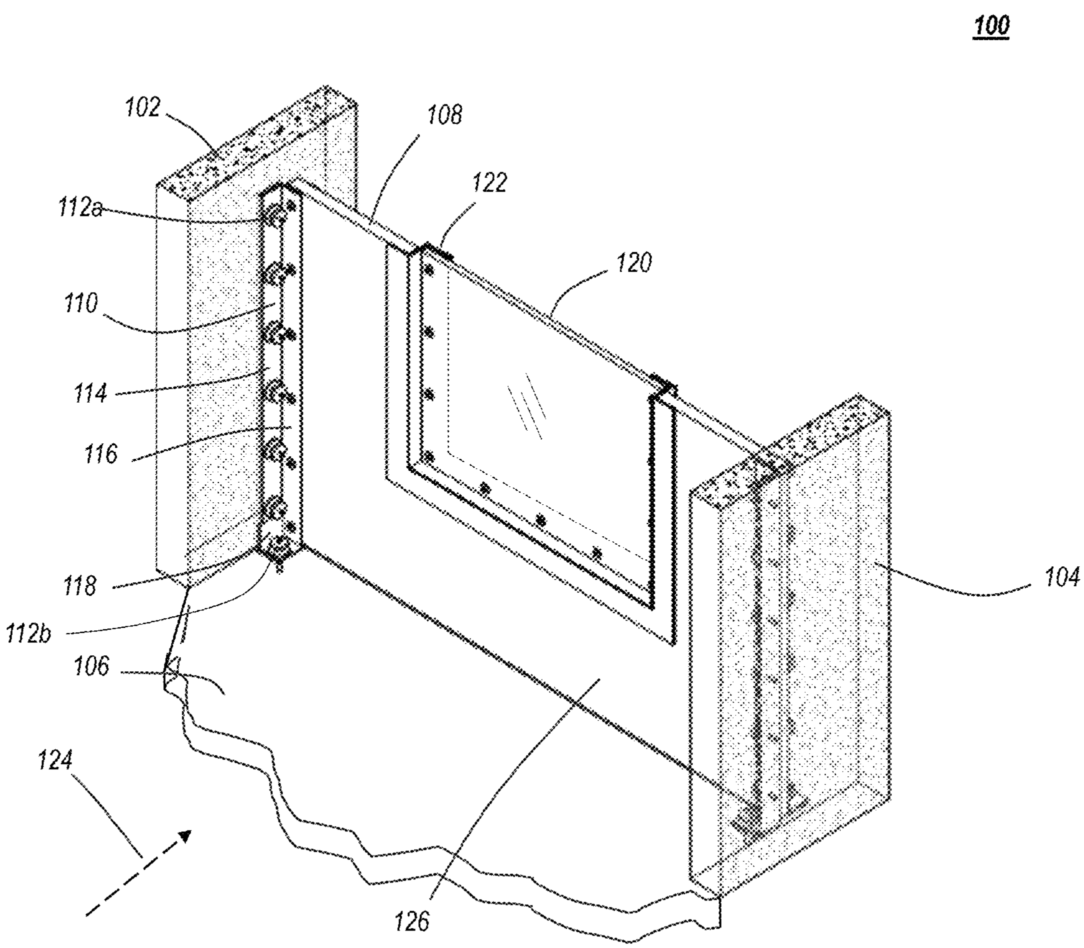


FIG. 1A

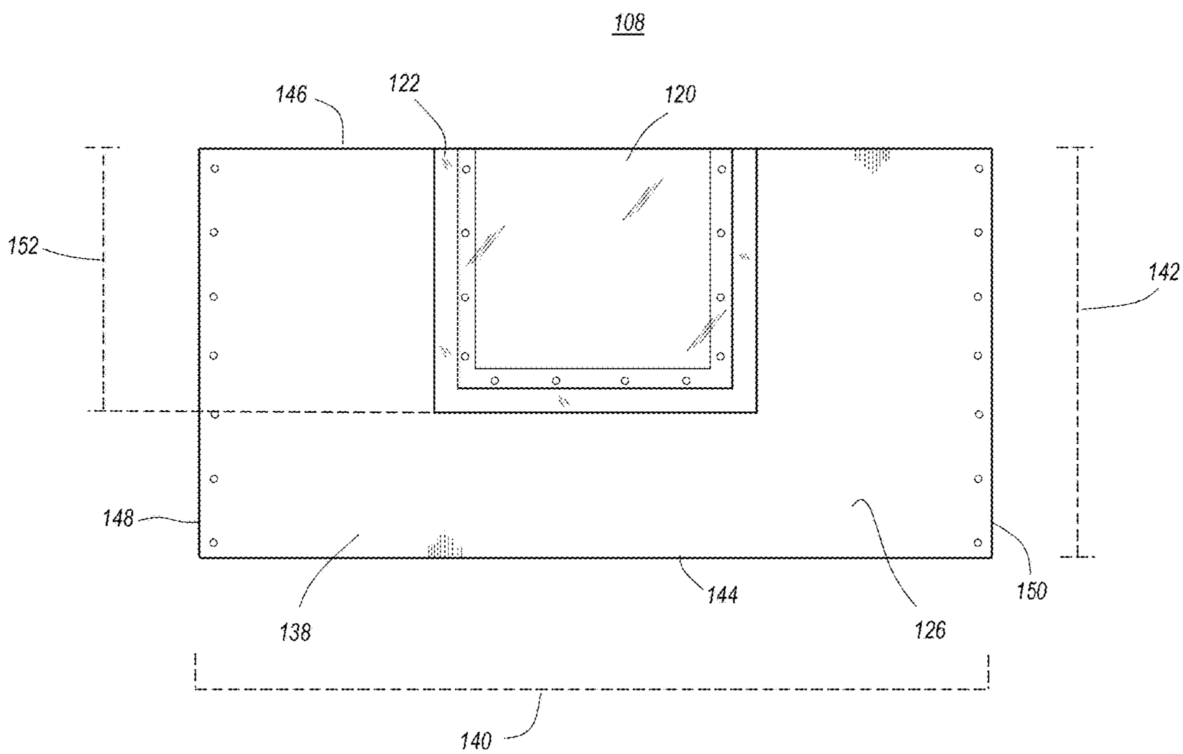


FIG. 1B

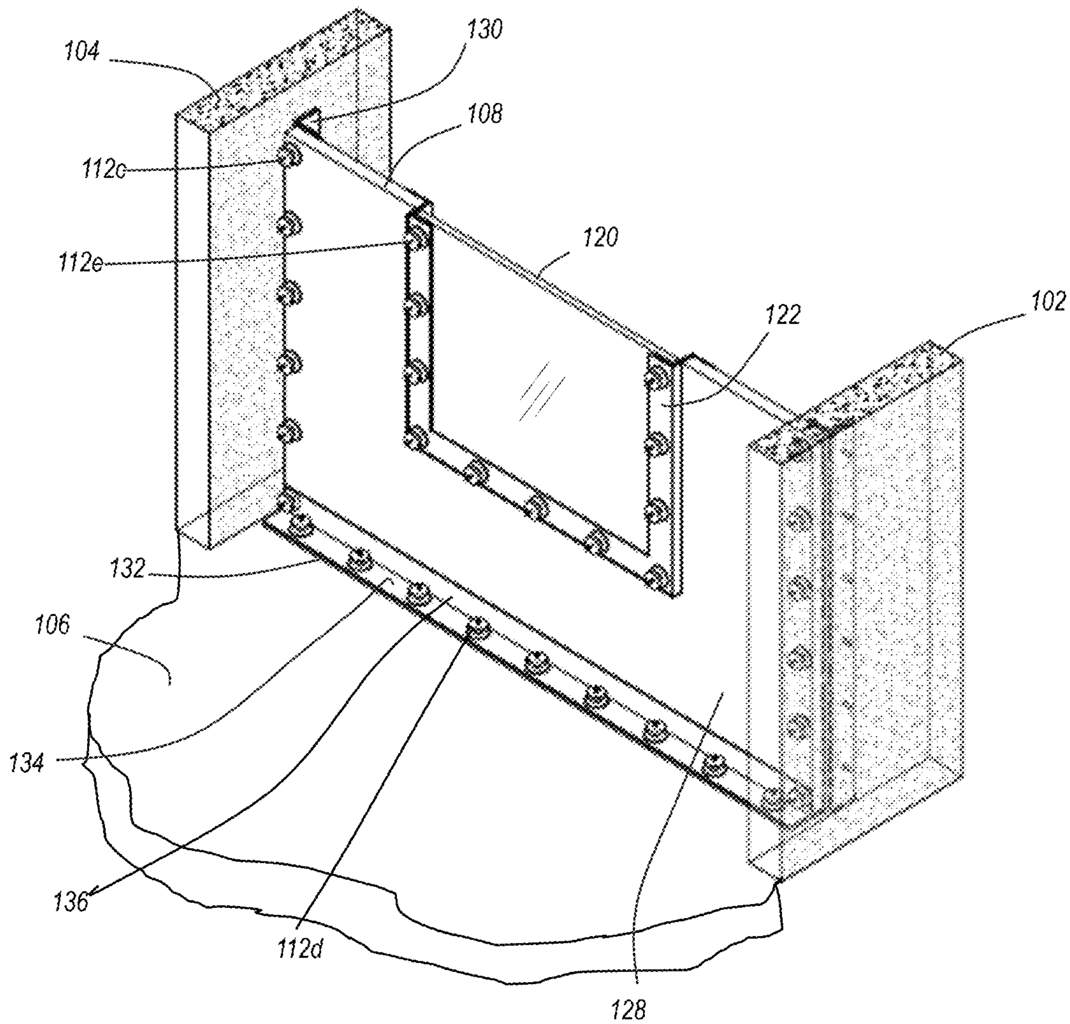


FIG. 2

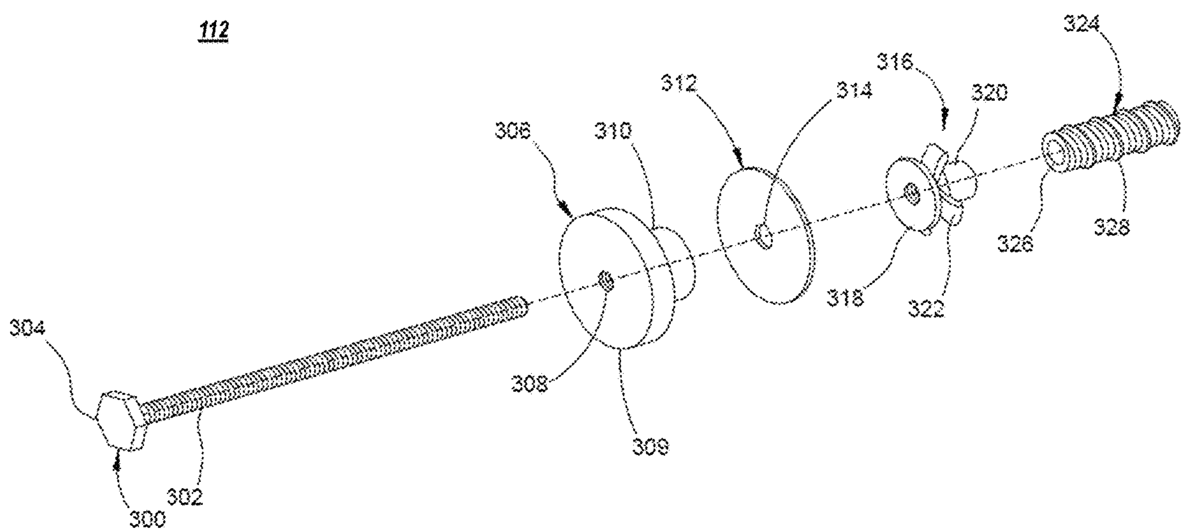

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FIG.3

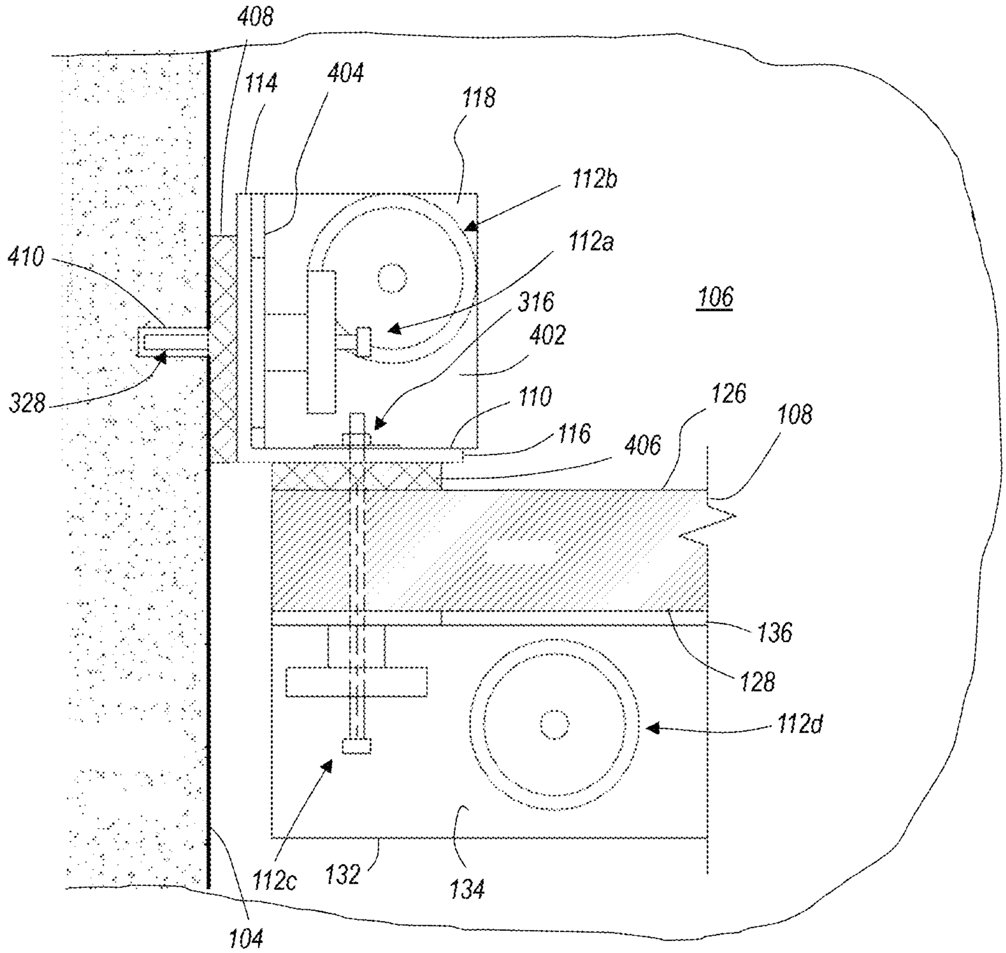
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FIG. 4

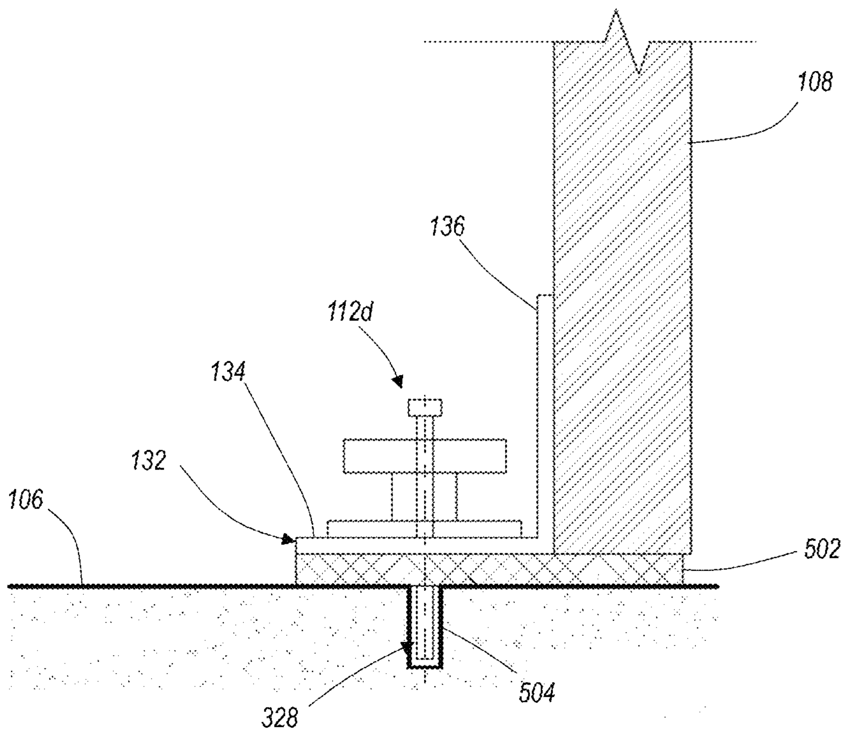

FIG. 5A

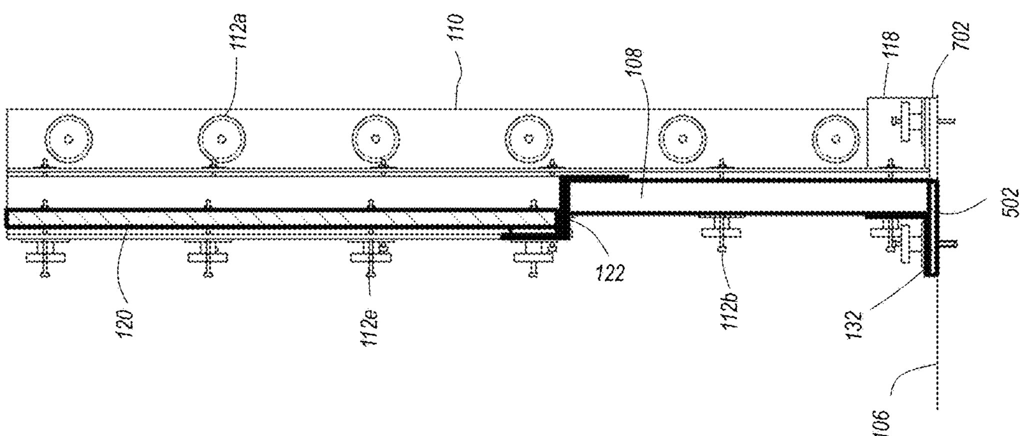

FIG. 5B

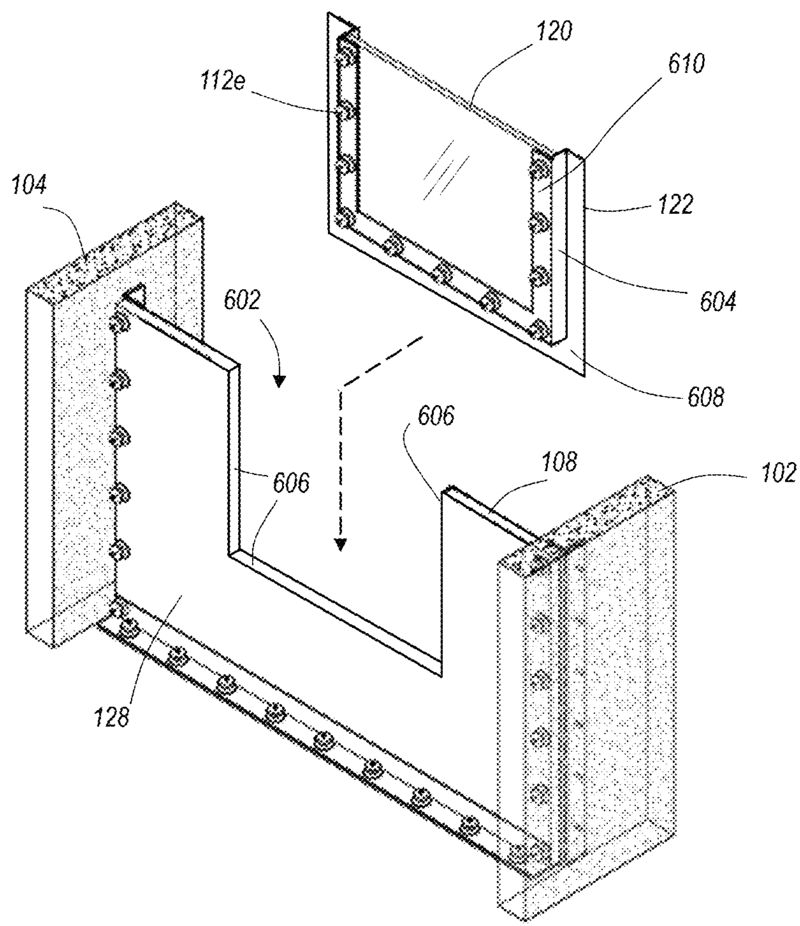

FIG. 6

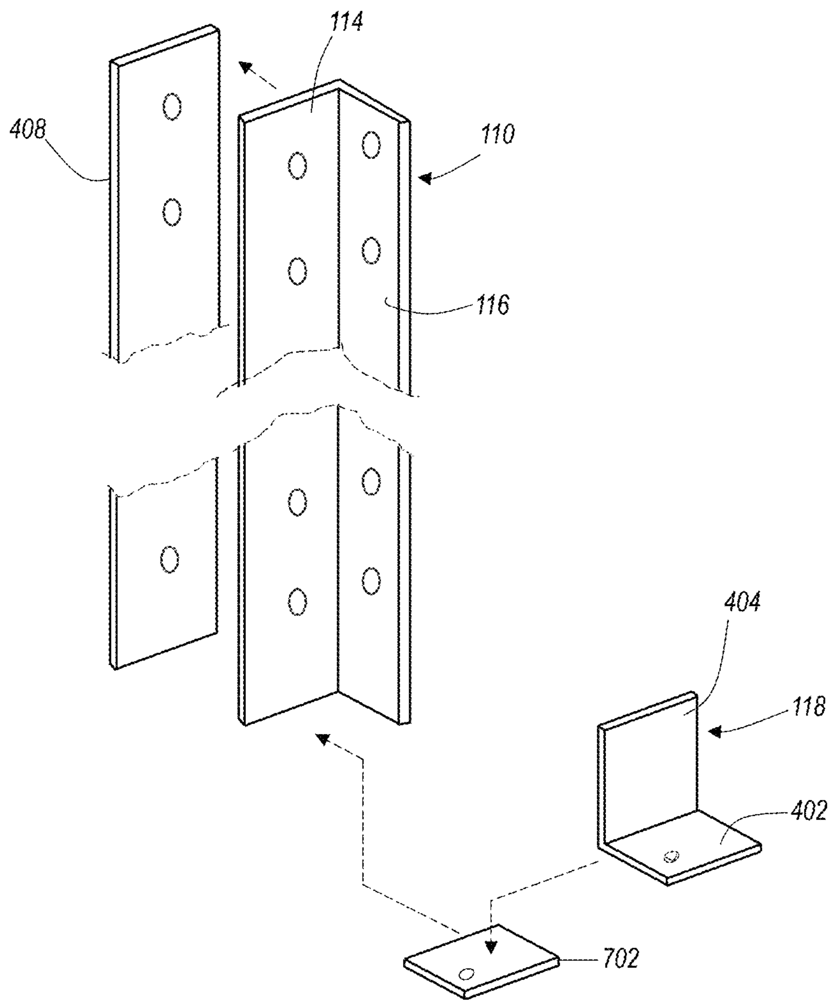

FIG. 7

FLOOD PANEL BARRIER SYSTEM FOR A THROUGHWAY OF A STRUCTURE

FIELD OF THE INVENTION

The present invention relates generally to flood barriers to prevent flood water intrusion into a building or other structure, and, more particularly, relates to a flood panel barrier system for a throughway, such as an alley or a doorway, or other space between two opposing walls.

BACKGROUND OF THE INVENTION

Water intrusion into buildings and structures due to flooding is responsible for a large amount of property damage. Flooding occurs as a result of rain, snow melt, and storm surge along coastlines, among other causes. As a result, flooding can occur in many different places in both rural and urban locations. In general, barriers have proven to be very effective at protecting property from flood water intrusion, and allow personnel to manage flood waters. There are numerous kinds of barriers, from sand bag barriers to earthen dykes with mechanically stabilized soil, and in many places panel systems are used. A panel system is a series of water-impermeable panels that are mounted to, or adjacent to a building or structure, and to the ground surface around the building or structure, in a substantially water tight arrangement. These barrier systems typically require angled braces for structural support on the back side (facing the building or structure being protected) to ensure the panels are not pushed inward and defeated by the water pressure during a flood event. This results in a minimum distance being necessary between the back of the panel barrier and the structure in order to fit the braces in between the panels and the building or structure. However, in some application, a building or structure need not be surrounded by a panel barrier because the external walls of the structure are effective at blocking water intrusion. This is true, for example, when the walls are made of poured concrete, for example. In those cases, only the throughways present an ingress point where water could intrude into the structure. A throughway is an opening into the structure that people pass through to enter and exit the structure, and there are typically opposing walls on each side of the opening. These walls are often part of the water-resistant portion of the structure. Conventionally, a flood panel barrier across a throughway may require a supporting member at the backside of the barrier (the dry side) to resist the force of water. In addition, if someone needed to ingress into, or egress from the structure through the throughway, they would have to climb over the panel barrier to do so, once it is in place.

Therefore, a need exists to overcome the problems with the prior art as discussed above.

SUMMARY OF THE INVENTION

In accordance with some embodiments of the inventive disclosure, there is provided a flood panel barrier system for a throughway. The system includes a flood panel that has a first side and a second side, and which extends across a throughway, from a first wall that defines a first side of the throughway, to a second wall that is opposite the first wall across the throughway and that defines a second side of the throughway. The flood panel is on a ground surface of the throughway at a bottom of the flood panel. The is a first vertical angled bracket that has a wall portion and a panel portion at an angle to each other. The wall portion is fastened

to the first wall. The panel portion is fastened to the flood panel at the first side of the flood panel. There is a first wall gasket disposed between the wall portion and the first wall, and a first panel gasket disposed between the panel portion and the first side of the flood panel. The system also includes a second vertical angled bracket that has a wall portion and a panel portion at an angle to each other. The wall portion is fastened to the second wall, and the panel portion is fastened to the flood panel at the first side of the flood panel. A second wall gasket is disposed between the wall portion and the second wall, and a second panel gasket disposed between the panel portion and the first side of the flood panel. The system further includes a horizontal angled bracket having a ground portion and a panel portion at an angle to each other. The ground portion is fastened to the ground surface between the first wall and the second wall and there is a floor gasket between the ground portion and the ground surface. The panel portion is fastened to the flood panel along the flood panel adjacent the bottom of the flood panel at the second side of the flood panel.

In accordance with a further feature, the flood panel includes a removable portion that extends from a top of the flood panel downward.

In accordance with a further feature, the removable portion include a frame that is sized to fit into a bay in the flood panel.

In accordance with a further feature, the wall portion of the first vertical angled bracket is fastened to the first wall by a first plurality of fastening units, the wall portion of the second vertical angled bracket is fastened to the second wall by a second plurality of fastening units, wherein each fastening unit of the first and second plurality of fastening units includes a bolt, an anchor embedded in a bore into which a distal end of the bolt is threaded, a washer, and an adjustment knob threaded onto the bolt.

In accordance with a further feature, there is further included, at the bottom of the first vertical angled bracket, and first ground support bracket, and at the bottom of the second vertical angled bracket, a second ground support bracket.

In accordance with some embodiments of the inventive disclosure, there is provided a flood panel barrier system for a throughway between a first wall and a second wall. The system includes a flood panel having a first or front side and a second or rear side. The flood panel extends across the throughway from the first wall to the second wall. A bottom or bottom edge of the flood panel is against a ground surface of the throughway, which can include a gasket between the bottom of the flood panel and the ground surface. The flood panel has a bay extending from a top of the flood panel downward toward the bottom of the flood panel. There is a removable portion mounted in a frame that is configured to fit into the bay in a water tight manner. There is a first vertical angled bracket having a wall portion and a panel portion at an angle to each other. The wall portion is fastened to the first wall, and the panel portion is fastened to the flood panel at the first side of the flood panel. There is a first wall gasket disposed between the wall portion and the first wall, and there is a first panel gasket disposed between the panel portion and the first side of the flood panel. There is a second vertical angled bracket having a wall portion and a panel portion at an angle to each other. The wall portion of the second angled bracket being fastened to the second wall, and the panel portion of the second angled bracket being fastened to the flood panel at the first side of the flood panel. There is a second wall gasket disposed between the wall portion and the second wall, and there is a second panel gasket disposed between the panel portion and the first side of the flood panel. The system also includes a horizontal angled bracket that has a ground portion and a panel portion at an angle to each other. The ground portion is fastened to the ground surface between the first wall and the second wall, and there is a floor gasket between the ground portion and the ground surface. The panel portion is fastened to the flood panel barrier along the flood panel adjacent the bottom of the flood panel at the second side of the flood panel.

In accordance with a further feature, the frame has a backing flange portion to which the removeable portion is fastened, and a front flange portion that extends outward to bear against a front side of the flood panel.

In accordance with a further feature, the wall portion of the first vertical angled bracket is fastened to the first wall by a first plurality of fastening units, the wall portion of the second vertical angled bracket is fastened to the second wall by a second plurality of fastening units, wherein each fastening unit of the first and second plurality of fastening units includes a bolt, an anchor embedded in a bore into which a distal end of the bolt is threaded, a washer, and an adjustment knob threaded onto the bolt.

In accordance with a further feature, the removable portion is transparent.

In accordance with a further feature, there is further included, at the bottom of the first vertical angled bracket, and first ground support bracket, and at the bottom of the second vertical angled bracket, a second ground support bracket.

In accordance with some embodiments of the inventive disclosure, there is also provided a flood panel for a flood panel barrier system that includes a planar body. The planar body has a back side and a front side, a width that extends from a first vertical edge to a second vertical edge, and a height that extends from a bottom to a top. The flood panel also includes a removable portion that fits into a bay in the planar body, and which extends from the top of the planar body downward. The removable portion is mounted in a frame that fits into the bay and creates a seal around the frame.

In accordance with a further feature, the frame comprises a backing flange portion that partially extends across a back side of the removable portion, a transverse portion that extends at a right angle from the backing flange portion a distance equal to a thickness of the planar body, and a front flange portion that extends outward from the transverse portion at a right angle to the transverse portion, opposite the distance from the backing flange portion.

In accordance with a further feature, the removable portion is fastened to the backing flange portion of the frame.

In accordance with a further feature, the removable portion is transparent.

In accordance with a further feature, the removable portion extends downward more than half of the height of the planar body.

Although the invention is illustrated and described herein as embodied in a flood panel barrier system, it is, nevertheless, not intended to be limited to the details shown because various modifications and structural changes may be made therein without departing from the spirit of the invention and within the scope and range of equivalents of the claims. Additionally, well-known elements of exemplary embodiments of the invention will not be described in detail or will be omitted so as not to obscure the relevant details of the invention.

Other features that are considered as characteristic for the invention are set forth in the appended claims. As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one of ordinary skill in the art to variously employ the present invention in virtually any appropriately detailed structure. Further, the terms and phrases used herein are not intended to be limiting; but rather, to provide an understandable description of the invention. While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward. The figures of the drawings are not drawn to scale.

Before the present invention is disclosed and described, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. The terms "a" or "an," as used herein, are defined as one or more than one. The term "plurality," as used herein, is defined as two or more than two. The term "another," as used herein, is defined as at least a second or more. The terms "including" and/or "having," as used herein, are defined as comprising (i.e., open language). The term "coupled," as used herein, is defined as connected, although not necessarily directly, and not necessarily mechanically. The term "providing" is defined herein in its broadest sense, e.g., bringing/coming into physical existence, making available, and/or supplying to someone or something, in whole or in multiple parts at once or over a period of time.

"In the description of the embodiments of the present invention, unless otherwise specified, azimuth or positional relationships indicated by terms such as "up", "down", "left", "right", "inside", "outside", "front", "back", "head", "tail" and so on, are azimuth or positional relationships based on the drawings, which are only to facilitate description of the embodiments of the present invention and simplify the description, but not to indicate or imply that the devices or components must have a specific azimuth, or be constructed or operated in the specific azimuth, which thus cannot be understood as a limitation to the embodiments of the present invention. Furthermore, terms such as "first", "second", "third" and so on are only used for descriptive purposes, and cannot be construed as indicating or implying relative importance.

In the description of the embodiments of the present invention, it should be noted that, unless otherwise clearly defined and limited, terms such as "installed", "coupled", "connected" should be broadly interpreted, for example, it may be fixedly connected, or may be detachably connected, or integrally connected; it may be mechanically connected, or may be electrically connected; it may be directly connected, or may be indirectly connected via an intermediate medium. As used herein, the terms "about" or "approximately" apply to all numeric values, whether or not explicitly indicated. These terms generally refer to a range of numbers that one of skill in the art would consider equivalent to the recited values (i.e., having the same function or result). In many instances these terms may include numbers that are rounded to the nearest significant figure. In this document, the term "longitudinal" should be understood to mean in a direction corresponding to an elongated direction of the article being referenced. Those skilled in the art can understand the specific meanings of the above-mentioned terms in the embodiments of the present invention according to the specific circumstances.

Conjunctive language such as the phrase "at least one of X, Y, and Z," unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require at least one of X, at least one of Y, and at least one of Z to each be present.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views and which together with the detailed description below are incorporated in and form part of the specification, serve to further illustrate various embodiments and explain various principles and advantages all in accordance with the present invention.

FIG. **1**A is a front perspective view of a flood panel barrier system for a throughway, in accordance with some embodiments.

FIG. **1**B is a front elevation view of a flood panel for a flood panel barrier system, in accordance with some embodiments.

FIG. **2** is a rear perspective view of a flood panel barrier system for a throughway, in accordance with some embodiments.

FIG. **3** is an exploded view of a fastener unit for use with a flood panel barrier system, in accordance with some embodiments.

FIG. **4** is a top sectional view of a flood panel barrier system where the flood panel meets one of the opposing walls of the structure, in accordance with some embodiments.

FIG. **5**A is a side sectional view of a flood panel barrier system, showing the bottom of the panel, in accordance with some embodiments.

FIG. **5**B is a side sectional view of a flood panel barrier system, in accordance with some embodiments.

FIG. **6** is a rear perspective view of a flood panel barrier system for a throughway in which a portion of the panel barrier can be removed to facilitate ingress into, or egress from the structure, in accordance with some embodiments.

FIG. **7** shows a detail of a vertical angled bracket of fastening a flood panel to a wall, in accordance with some embodiments.

DETAILED DESCRIPTION

While the specification concludes with claims defining the features of the invention that are regarded as novel, it is believed that the invention will be better understood from a consideration of the following description in conjunction with the drawing figures, in which like reference numerals are carried forward. It is to be understood that the disclosed embodiments are merely exemplary of the invention, which can be embodied in various forms.

FIG. **1**A is a front perspective view of a flood panel barrier system **100** for a throughway, in accordance with some embodiments. FIG. **2** is a rear perspective view of the flood panel barrier system **100**, and reference should be made to both drawings in the following description. FIG. **1** shows the "wet" side of the barrier with flood water coming from the direction of arrow **124**. A throughway is defined between two opposing walls **102**, **104**, where only a section of the walls **102**, **104** are shown there it should be understood that the walls **102**, **104** extend upwards, as well as to the front and rear. There is a ground surface **106** between the walls **102**, **104** that can be a water-impermeable surface, as are the walls **102**, **104**. The walls **102**, **104** and ground surface **106** can define an entryway into a building or other structure, or it can be a separation between two adjacent structures, such as, for example, an alley or walkway. The throughway is a point at which flood water could pass, and enter into a structure or structures.

The prevent flood water ingress through the throughway, a flood panel **108** is placed between, and fastened to the walls **102**, **104** in a "water tight" arrangement that prevents flood waters from passing through the throughway and into the structure. The flood panel, or simply "panel" **108** is a rigid, water impermeable member having a generally planar body that can be sized for a specific throughway, or standardized to fit on standardized throughways. The planar body of the flood panel **108** has a width, a height, and a thickness. The width is measured in the horizontal direction when the panel **108** is placed in the throughway. The height is measured in the vertical direction, and the thickness is measured from the first or front side to the second or rear side in a direction perpendicular to the sides of the planar body. The planar body also has a first vertical edge. The panel **108** has a front side **126** (FIG. **1**) and a rear side **128** (FIG. **2**), which can be defined by the internal structure of the panel **108**, or simply by how it is placed in the throughway (e.g., the "front" side is the one facing outward from the throughway when the panel **108** is put in place in the throughway). At the ends of the panel **108**, which are against, or very close to the walls **102**, **104**, and at the front side **126**, there are vertical angled brackets **110**, **130** that fasten the panel **108** to the walls **102**, **104**. The vertical angled brackets **110** have a wall portion **114** that interfaces with the wall (**102** or **104**), and a panel portion **116** that interfaces with the panel **108** at the front side **126** of the panel **108**. The wall portion **114** and panel portion **116** can be at a right angle to each other, and extend from the ground surface **106** upwards. The ground surface **106** is understood to be a hard, water impermeable material, such as, for example, concrete. A plurality of fastening units **112*a*** include a bolt that passes through the wall portion **114** and into an anchor embedded in a corresponding bore in the wall **102**, **104**, as will be described. The fastening units described herein all use the reference numeral "**112**," and have an alphabetic suffix a-e to indicate their location, but they are otherwise very similar, as will be described in reference to FIG. **3**. The ground surface **106** must both resist erosive forces of flowing and standing water, as well as provide a suitable anchor medium for fastening units **112**. The fastening units **112*a*** are disposed at regular intervals along the wall portion **114** and fasten the wall portion **114** to the wall **102**, **104**. In addition, there are other fastening units **112*c*** that pass through the panel **108** and the panel portion **116** to fasten the panel portion **116** to the panel **108**. The vertical angled brackets **110**, **130** are made of a rigid material, such as any of various polymeric materials, aluminum, stainless steel, and so on. There are openings at regular intervals along the length of the angled vertical brackets **110**, **130** in both the wall portion **114** and the panel portion **116** for the fastening units **112*a***, **112*c***.

At the bottom of the vertical angled brackets **110**, **130** there is a ground support bracket **118** that is itself an angled bracket having a vertical portion that is against the wall portion **114**, and a horizontal portion that interfaces against the ground surface **106**, and which is held in place by a fastening unit **112**b that connects to an anchor that is embedded in a bore in the ground surface **106**. The ground support bracket **118** also has an inner side along both the horizontal and vertical portions that bear against the panel portion **116** of the vertical angled bracket **110**.

FIG. **1**B shows a front elevation view of a flood panel **108** for a flood panel barrier system **100**, in accordance with some embodiments. The flood panel **108** includes a generally planar body **138** having a front side **126** and a rear side (**128**). The planar body **138** has a width **140** which is measured in the horizontal direction when the panel is oriented and placed for use. It has a height **142**, and the removable portion **120** and frame can extend from the top **146** downwards, towards the bottom **144** of the planar body **138** a distance **152** that is more than half the height **142** in some embodiments. The planar body **138** also has a first vertical edge **148** and a second vertical edge **150**.

At the rear side **128**, as seen in FIG. **2**, there is a horizontal angled bracket **132** that is used to fasten the bottom of the panel **108** to the ground surface **106**. The horizontal angled bracket **132** extends across the entire width of the panel **108**, from the first wall **102** to the second wall **104**, and has a ground portion **134** that interfaces with the ground surface **106**, and a panel portion **136** that interface with the panel **108** at the bottom of the panel **108** adjacent the bottom edge of the panel **108**. A plurality of fastening units **112**d fasten the ground portion **134** of the horizontal angled bracket **132** to the ground surface **106**. Each one of the fastening units **112**d includes a bolt that passes through the ground portion **134** and into an anchor that is embedded in a bore in the ground surface **106**. Another plurality of fastening units **112**c passes through the panel **108** at the sides, in a vertical series, to fasten the panel portion **116** of the vertical angled brackets **110**, **130**. As can be seen in FIG. **2**, the tightening portion of the fastening units **112**c are on the rear side **128** of the panel **108**. This allows personnel to tighten any of the fastening units **112**c from inside the structure, when there is flood water present at the front side **126** of the panel **108**. Likewise, having the horizontal angled bracket **132**, with its fastening units **112**d on the "dry" side of the panel **108** allows tightening of the fastening units **112**d as may be needed, when there is flood water present at the front side of the panel **108**.

The flood panel **108** can be on the order of 48" to 60" tall, from the ground edge to the top edge. It can be taller or shorter in some applications. The vertical angled brackets **110**, **130** can extend the full height of the flood panel **108**. Prior to the arrival of flood waters, with the flood panel barrier system **100** in place, it may be desirable to continue using the throughway to enter or exit the structure. However, at four to five feet tall, it is not possible without climbing over the flood panel **108** and possibly damaging the flood panel **108** and/or loosening the fastening units **112**. To facilitate passing through the throughway with the flood panel barrier system **100** in place, the flood panel **108** includes a removable portion **120** that is held in a frame **122** that fits into a corresponding bay **602** of the panel **108**. Reference is further made to FIG. **6** here in describing the removable portion **120** and frame **122**. The removable panel **120** can be made of a transparent material; this will allow personnel inside the structure to visually gauge the height of flood water once it rises to a level equal to the bottom of the bay **602**. The bay **602** can be a rectangular "cutout" at the top of the panel **108** and extending downward. In some embodiments the bay **602** can be on the order of four feet wide and three feet in height. That means that the removable portion **120**, and in particular the bay **602**, can extend downward, from the top of the flood panel **108**, more than half the height of the flood panel **108**. The bay can be larger or smaller some applications. With the removable portion **120** and frame **122** removed from the bay **602**, a person can step over the center portion of the panel **108** through the bay **602** to exit or enter the structure. The removable portion **120** is fastened to a backing flange portion **610** of the frame **122** that is generally planar, and, when the frame is positioned properly in the bay **602**, is co-planar with the rear side **128** of the flood panel **108**. A plurality of fastening units **112**e hold the removable portion **120** in the frame **122** against the backing flange portion **610**, passing through the backing flange portion **610** and the removable portion **120** adjacent the edges of the removable portion **120**. The frame **120** has a transverse portion **604** at a right angle to the backing flange portion **610** which mates with the edge **606** of the panel **108** around the bay **602**, and which extends across the thickness of the planar body of the flood panel **108**. The transverse portion **604** also mates with the removeable portion **120** on the opposite side of the transverse portion **604** from the side that mates with the edge **606**. A front flange **608** or front flange portion extends outward from the transverse portion **604**, at a right angle, and is configured to rest against the front surface **126** of the panel **108**. There can be a layer of compliant material disposed on the side of the front flange **608** to act as a gasket. When the frame **122** is placed in the bay **602**, the frame **122** is sized so that there is contact between the edge **606** and the transverse portion **604** of the frame around the entirety of the transverse portion **604**, but the fit is not so tight as to prevent removal of the panel **122** from the bay **602**. With the frame **122** in the bay **602**, hydrostatic pressure of flood water, when present, at the front side of the panel **108** will seal the front flange **608** against the front surface **126** of the panel **108**.

FIG. **3** shows an exploded view of a fastener unit **112** for use with the inventive flood panel barrier system support structure, in accordance with some embodiments. The fastener unit **112** can be any of the fastener units **112**a-**112**e shown in the other drawings. Each fastener unit **112** includes a bolt **300** which can be a tap bolt. The bolt **300** includes a threaded shank **302** and a head **304** that can be a hex-head, as is well known. An adjustment knob **306** has a threaded through-hole **308** to allow the adjustment knob **306** to thread onto the shank **302** of the tap bolt **300**. The main body **309** is puck-shaped and can be knurled or fluted around the outside edge to allow for turning the knob **306**. The knob **306** further includes a standoff **310** that can also be puck-shaped, with a smaller diameter than the main body **309**, and through which the threaded hole **308** also passes. The standoff **310** allows some space between the main body **309** and the washer **312** to allow a person to get better purchase on the main body **309** when turning the main body **309** and knob **306**. The bottom of the standoff portion **310** of the knob **306** bears against the washer **312** to exert a compressive force, and the washer **312** has an opening **314** that is slightly larger than the outside diameter of the threads on the shank **302** of the bolt. Thus, the shank **302** can pass through the washer **312** without having to be threaded through the washer **312**. The washer **312** can be a dock washer and have a diameter of about three inches in some embodiments. The bolt **300** can have a length of four to six inches, such as a ⅜"×4" tap bolt.

The distal end of the shank **302** threads into either a rivet nut **316** or an anchor **324**, depending on the location of the fastening unit **112**. The rivet nut **316** includes a threaded body **320** into which the shank **302** is threaded. A flange **318** extends outward and acts like a washer. There is a collapsible portion **322** that collapses when the bolt is tightened while the flange **318** is bearing against a surface, causing sections of the collapsible portion to collapse and extend outward. The rivet nut is used in fastening units **112***c* and **112***e*. The other fastening units **112***a*, **112***b*, and **112***d* use the anchor **324**.

The anchor **324** can be a snake anchor and is intended to fit into a bore, such as a bore into the walls **102**, **104** or the ground surface **106**, and frictionally engage the sides of the bore so as to become stuck fast in the bore. This can occur, for example, by making the inner diameter of the axial bore **326** of the anchor **324** slightly smaller than the outside diameter of the shank **302** of the bolt **300**, and the outside diameter of the body **328** of the anchor **324** is made to be about the same as the diameter of the bore into which the anchor **324** is sunk (e.g. in the walls **102**, **104** or ground surface **106**). Thus, with the anchor **324** placed into a bore in the walls or ground surface, when the bolt **300** is threaded into the axial bore **326** of the anchor, it will force the body **328** outward against the wall of the bore in the wall or ground surface in which the anchor **324** is sunk.

FIG. **4** is a top sectional view of a flood panel barrier system where the flood panel **108** meets one of the opposing walls **104** of the structure, in accordance with some embodiments. The vertical angled bracket **110** is shown with the wall portion **114** and panel portion **116** at a right angle to each other. A wall gasket **408** is compressed between the wall portion **114** and the surface of the wall **104** by fastening unit **112***a*, which has a bolt that threads into an anchor **328** in a bore **410** in the wall **104**. A panel gasket **406** is disposed between the panel portion **116** and the front surface **126** of the panel **108**, and is compressed by fastening unit **112***c*, which uses a rivet nut **316** rather than an anchor (e.g. **328**). At the bottom of the vertical angled bracket **110** is the ground support bracket **118**, which has a horizontal portion **402** that bears against the ground surface **106**, and a vertical portion **404** that bears against the wall portion **114** of the vertical angled bracket **110**. Fastening unit **112***b* fastens the ground support bracket **118** to the ground surface **106**.

At the rear side **128** of the panel, there is shown the horizontal angled bracket **132**, which has a ground portion **134** that in co-planar with the ground surface **106**, and fastened to the ground surface by fastening units **112***d*, compressing a ground gasket **502** (as shown in FIG. **5**). The panel portion **136** of the horizontal angled bracket **132** bears against the panel **108** at the rear side **128** of the panel **108**.

FIG. **5**A is a side sectional view of a flood panel barrier system, in accordance with some embodiments. The section is taken through the panel **108**, horizontal angled bracket **132**, and floor gasket **502**. The floor gasket **502** extends under both the panel **108** and the ground portion **134** of the horizontal angled bracket **132**. Fastening units **112***d* fasten the horizontal angled bracket **132** to the ground surface **106**. A bore **504** in the ground surface **106** has an anchor **328** embedded in it, which the bolt of the fastening unit **112***d* thread into, causing the anchor to expand outward against the sides of the bore **504**. The panel **108** is held in place against the floor gasket by the vertical angled brackets **110**, which are not shown in this drawing. FIG. **5**B shows a side sectional view of a flood panel barrier system, in which the vertical angled bracket **110** is shown, along with the ground support bracket **118**. There is a second floor gasket **702**, or an extension of the floor gasket **502**, under the ground support bracket **118**, which is fastened to the ground surface.

FIG. **7** shows a detail of a vertical angled bracket **110** for fastening a flood panel to a wall, in accordance with some embodiments. The wall portion **114** bears against the wall via a wall gasket **408**. The panel portion **116** bears against a panel (not shown). A ground support bracket **118** is shown which bears against the bottom of the vertical angled bracket **110** such that the vertical portion **404** is against the surface of the wall portion **114** of the vertical angled bracket **110**, and the horizontal portion **402** is fastened to the ground surface, compressing a floor gasket **702**.

A flood panel barrier system has been disclosed that is useful for preventing flood water intrusion into a structure at a throughway of the structure, or between structures. The flood panel barrier system fastens a flood panel between, and to opposing walls, and the ground surface, in a manner that is water tight. This allows the structure itself to be the primary barrier to flood waters, assuming the walls of the structure are water impermeable. The disclosed flood panel can have a removable portion to allow persons to pass through the panel by stepping over the lower portion of the panel and through a bay in which the removeable portion is otherwise placed.

The claims appended hereto are meant to cover all modifications and changes within the scope and spirit of the present invention.

What is claimed is:

1. A flood panel barrier system for a throughway, comprising:

a flood panel having a front side and a rear side, the flood panel extending across a throughway, from a first wall that defines a first side of the throughway to a second wall that is opposite the first wall across the throughway and that defines a second side of the throughway, the flood panel further having a bottom that is against a ground surface, and a top opposite the bottom, a bay formed in the flood panel from the top downward into the flood panel toward the bottom of the flood panel;

a first vertical angled bracket having a wall portion and a panel portion at an angle to each other, the wall portion being fastened to the first wall, the panel portion being fastened to the flood panel at the front side of the flood panel, a first wall gasket disposed between the wall portion and the first wall, and a first panel gasket disposed between the panel portion and the front side of the flood panel;

a second vertical angled bracket having a wall portion and a panel portion at an angle to each other, the wall portion being fastened to the second wall, the panel portion being fastened to the flood panel at the front side of the flood panel, a second wall gasket disposed between the wall portion and the second wall, and a second panel gasket disposed between the panel portion and the front side of the flood panel;

a horizontal angled bracket having a ground portion and a panel portion at an angle to each other, the ground portion being fastened to the ground surface between the first wall and the second wall and having a floor gasket between the ground portion and the ground surface, the panel portion being fastened to the flood panel along the flood panel adjacent the bottom of the flood panel at the rear side of the flood panel; and

a removable portion mounted in a frame that is configured to fit into the bay, wherein the removable portion is transparent, and wherein the frame is configured to be in a watertight contact with the frame and creates a seal around the frame.

2. The flood panel barrier system of claim **1**, wherein the frame of the removable portion includes:

a backing flange portion that is co-planar with the rear side of the flood panel when the removable portion is positioned in the bay;

a transverse portion formed at a right angle to the backing flange portion that extends from the backing flange portion towards the front side of the flood panel, across a thickness of the flood panel and which mates with an edge of the panel around the bay;

a front flange portion that extends outward from the transverse portion at a right angle at the front side of the flood panel when the removable portion is positioned in the bay, and is configured to rest against the front side of the flood panel; and

wherein the transparent member fits in the frame against the backing flange portion and the transverse portion.

3. The flood panel barrier system of claim **1**, wherein the bay extends downward into the flood panel more than half a height of the flood panel.

4. The flood panel barrier system of claim **1**, wherein the wall portion of the first vertical angled bracket is fastened to the first wall by a first plurality of fastening units, the wall portion of the second vertical angled bracket is fastened to the second wall by a second plurality of fastening units, wherein each fastening unit of the first and second plurality of fastening units includes a bolt, an anchor embedded in a bore into which a distal end of the bolt is threaded, a washer, and an adjustment knob threaded onto the bolt.

5. The flood panel barrier system of claim **1**, further including, at the bottom of the first vertical angled bracket, and first ground support bracket, and at the bottom of the second vertical angled bracket, a second ground support bracket.

6. A flood panel barrier system for a throughway between a first wall and a second wall, comprising:

a flood panel having a first side and a second side, the flood panel extending across the throughway from the first wall to the second wall, the flood panel further on a ground surface of the throughway at a bottom of the flood panel, the flood panel having a bay extending from a top of the flood panel downward;

a removable portion mounted a frame, the frame being sized to fit into the bay in a water tight manner;

a first vertical angled bracket having a wall portion and a panel portion at an angle to each other, the wall portion being fastened to the first wall, the panel portion being fastened to the flood panel at the first side of the flood panel, a first wall gasket disposed between the wall portion and the first wall, and a first panel gasket disposed between the panel portion and the first side of the flood panel;

a second vertical angled bracket having a wall portion and a panel portion at an angle to each other, the wall portion being fastened to the second wall, the panel portion being fastened to the flood panel at the first side of the flood panel, a second wall gasket disposed between the wall portion and the second wall, and a second panel gasket disposed between the panel portion and the first side of the flood panel; and

a horizontal angled bracket having a ground portion and a panel portion at an angle to each other, the ground portion being fastened to the ground surface between the first wall and the second wall and having a floor gasket between the ground portion and the ground surface, the panel portion being fastened to the flood panel barrier along the flood panel adjacent the bottom of the flood panel at the second side of the flood panel.

7. The flood panel barrier system of claim **6**, wherein the frame has a backing flange portion to which the removeable portion is fastened, and a front flange portion that extends outward to bear against a front side of the flood panel.

8. The flood panel barrier system of claim **6**, wherein the wall portion of the first vertical angled bracket is fastened to the first wall by a first plurality of fastening units, the wall portion of the second vertical angled bracket is fastened to the second wall by a second plurality of fastening units, wherein each fastening unit of the first and second plurality of fastening units includes a bolt, an anchor embedded in a bore into which a distal end of the bolt is threaded, a washer, and an adjustment knob threaded onto the bolt.

9. The flood panel barrier system of claim **6**, wherein the removable portion is transparent.

10. The flood panel barrier system of claim **6**, further including, at the bottom of the first vertical angled bracket, and first ground support bracket, and at the bottom of the second vertical angled bracket, a second ground support bracket.

11. A flood panel for a flood panel barrier system, comprising:

a planar body having a back side and a front side, a width that extends from a first vertical edge to a second vertical edge, and a height that extends from a bottom to a top; and

a removable portion that fits into a bay in the planar body and extends from the top of the planar body downward, and wherein the removable portion is mounted in a frame that fits into the bay and creates a seal around the frame.

12. The flood panel of claim **11**, wherein the frame comprises:

a backing flange portion that partially extends across a back side of the removable portion, a transverse portion that extends at a right angle from the backing flange portion a distance equal to a thickness of the planar body, and a front flange portion that extends outward from the transverse portion at a right angle to the transverse portion, opposite the distance from the backing flange portion.

13. The flood panel of claim **12**, wherein the removable portion is fastened to the backing flange portion of the frame.

14. The flood panel of claim **11**, wherein the removable portion is transparent.

15. The flood panel of claim **11**, wherein the removable portion extends downward more than half of the height of the planar body.

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